Bayesian Sequential Decision-Making

in Non-Stationary, Heavy-Tailed Environments

Honors Thesis Defense — May 2026

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The Sticky Prior Paradox — Motivation & Problem Statement

The Scenario

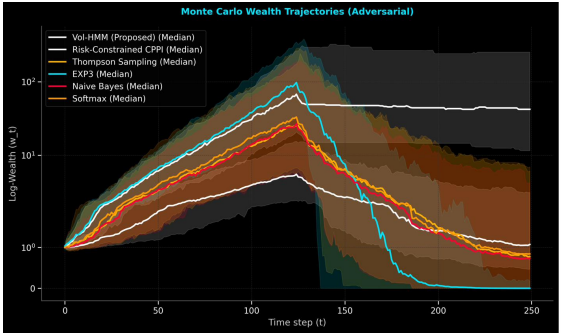
- Agent observes $T_0 = 200$ bull-market days. *Posterior*: 99% confidence in positive drift.
- At step 201: structural break to a heavy-tailed bear regime.

The Failure (Proposition 3.3)

- Detection lag $\approx \frac{T_0}{2|p_1 - p_0|} = O(T_0)$.
- Agent continues betting $f \approx f^*_{\text{bull}}$ into crash.
- **Wealth $\to 0$ within $\sim$20 steps.**

Three Contributions

C1 Rigorous simulation (CRN, strict filtration)



Monte Carlo fan ($N = 100$). Break at $T = 125$. **Vol-HMM** preserves capital; baselines collapse.

Mathematical Framework — Filtered Probability Space & Kelly–Breiman

Measure-Theoretic Setup

$$(\Omega,\ \mathcal{F},\ \{\mathcal{F}_t\}_{t=0}^{T},\ \mathbb{P})$$

- $\mathcal{F}_t = \sigma(r_1, \ldots, r_t)$ — info through t.
- **Admissibility**: $f_t \in \mathcal{F}_{t-1}$-measurable (decide *before* r_t is revealed).
- **Wealth**: $W_T = \prod_{t=1}^{T}(1 + f_t r_t)$, stored in log-space for numerical stability.

Kelly–Breiman Theorem

Let $f^* = \arg\max_f \mathbb{E}[\ln(1 + fr)]$. Then a.s.:

$$\frac{1}{T} \ln W_T^{(f^*)} \xrightarrow{a.s.} \mathbb{E}[\ln(1 + f^* r)]$$

For any $f \neq f^*$: $W_T^{(f^*)} / W_T^{(f)} \xrightarrow{a.s.} +\infty$

The Problem

The SLLN holds only when $r_1, r_2, \ldots$ are i.i.d. Market regime switches break this assumption. No amount of parameter tuning recovers it.

Sticky Prior (Proposition 3.3)

After T_0 Bull observations the Beta posterior needs $\approx \dfrac{T_0}{2|p_1 - p_0|}$ Bear samples to correct its mean.

For $T_0 = 200$, $|\Delta p| = 0.5$: ≈ 200 steps.

The lag is **structural** — it requires a change to the observation space to fix.

Agent Hierarchy — Baselines vs. Proposed Methods

Baseline Agents

- **Kelly Oracle** — knows true μ_t, σ_t^2. Theoretical upper bound; not implementable.
- **Naive Bayes Kelly** — plug-in posterior mean. Optimal IID; suffers $O(T_0)$ lag.
- **Thompson Sampling** — samples $\tilde{p} \sim \mathrm{Beta}(\alpha, \beta)$. Calibrated uncertainty; same structural lag.
- **Softmax** — temperature-weighted allocation.
- **EXP3** ($n = 1$ arm) — degenerates to fixed $f = 0.5$; cannot de-lever.

Proposed: Vol-HMM

2D observation (r_t, v_t) + CUSUM trigger. Rolling volatility spikes immediately on break; no prior mass to overcome.

Lag: 14.3 → 1.9 steps (≈87% reduction).

Proposed: Risk-Constrained CPPI

Mechanical leverage floor — regime-agnostic. Bounds drawdown even when inference is wrong.

0% empirical ruin ($N = 100$, adversarial).

Why both? HMM handles inference speed. CPPI handles tail risk. Neither alone suffices.

Proposed Method 1: Volatility-Augmented HMM — C2 — Fast Regime Detection

Why Add Volatility?

Bayesian posteriors update on return *sign* only. Rolling volatility $v_t = \mathrm{std}(r_{t-W+1:t})$ spikes within one step of a break — no prior mass accumulation.

Joint Emission (log-space)

$$\ln b_j(r_t, v_t) = \ln \mathcal{N}(r_t; \mu_j, \sigma_j^2) + \ln \Gamma(v_t; k_j, \theta_j)$$

Params specified a priori; EM would overfit at $T = 250$.

Forward Filter

$$\ln \alpha_t(j) = \ln b_j(y_t) + \mathrm{logsumexp}_i[\ln \alpha_{t-1}(i) + \ln A_{ij}]$$

$A_{ii} \leq 0.95$ prevents sticky-prior re-emergence.

CUSUM Trigger

$$S_t^- = \max(0, S_{t-1}^- - z_t - k), \quad h = 3.0.$$

Detection Lag vs. A_{ii} (Table 5.3)

A_{ii}	Lag	Ruin
0.99	14.3	32%
0.98	8.7	15%
0.95	**1.9**	**0%**
0.80	1.2	0%*

*Over-sensitivity causes whipsawing.

Key Result

$A_{ii} = 0.95$ optimal.
14.3 → 1.9 steps = **86.7% reduction**.

Cost of Safety Cap

Proposed Method 2: Risk-Constrained CPPI — C3 — Provable Drawdown Floor

Floor, Cushion, Leverage

$$\bar{W}_t = \max_{s \le t} W_s \quad \text{(running peak)}$$
$$F_t = (1 - D_{\max})\,\bar{W}_t \quad \text{(floor)}$$
$$C_t = W_t - F_t \quad \text{(cushion)}$$
$$\lambda_t = \min(1,\ m \cdot C_t / W_t)$$

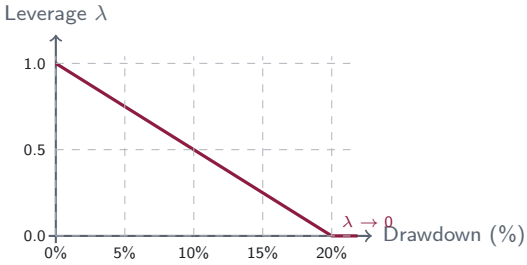
Parameters: $m = 3.0$, $D_{\max} = 20\%$.

Proposition 4.2

If $m \cdot |r_{\min}| \le 1$, then $W_{t+1} \ge F_t$.

Discrete-time caveat. Student-t_3 bear returns have $|r| > 1/3$ with prob. $\approx 24.7\%$/step. Guarantee is probabilistic in discrete time. *Empirical*: 0% ruin over $N = 100$ simulations.

Leverage vs. Drawdown from Peak



As wealth hits floor at $D_{\max} = 20\%$, leverage drops to zero (cash).

Complementarity

CPPI is regime-agnostic: it protects capital even when the HMM is wrong. HMM reduces

Cost of Survival

Adversarial: 1.07 vs. Vol-HMM 40.3

Results: Adversarial Structural Break — $N = 100$ simulations, $T = 250$, CRN enforced

Setup

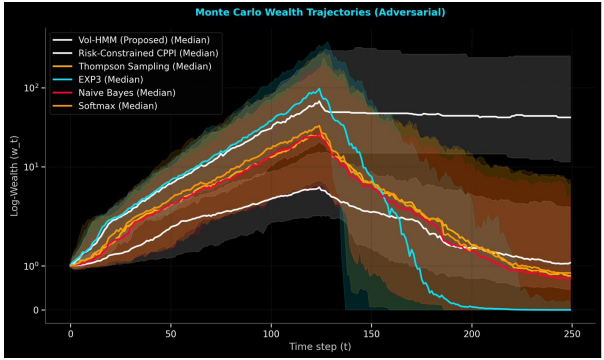
Bull: $r \sim \mathcal{N}(0.08, 0.15^2)$, $t \leq 125$.

Bear: $r \sim t_3(-0.10, 0.30^2)$, $t > 125$.

Ruin Risk

Agent	Ruin
EXP3 (fixed $f = 0.5$)	**43%**
Thompson Sampling	**16%**
Naive Bayes / Softmax	**12%**
Vol-HMM	**3%**
CPPI	**0%**†

†Empirically; discrete-time caveat applies.



Vol-HMM deleverages at $T = 125$ and diverges upward. EXP3 (blue): fixed $f = 0.5$ into a negative-drift heavy-tail regime.

CPPI (lower): capital preserved at the cost of growth.

Median Terminal Wealth

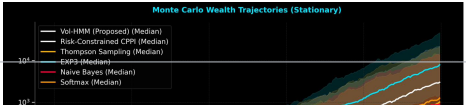
Vol-HMM: 42.28× **CPPI**: 1.07×

Results: Stationary & Slow Drift — Cost of caution made visible

Stationary ($\mu = 0.08$, $\sigma = 0.15$, $T = 250$)

Agent	Median W_{250}
EXP3	7,944×
Vol-HMM	3,019×
Thompson	1,012×
CPPI	34.9×

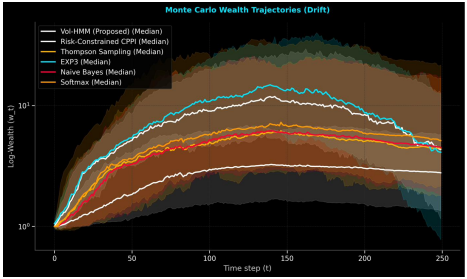
Vol-HMM lower than EXP3: $f \leq 0.5$ cap. This $\sim 2.6\times$ gap is the exact cost of the safety cap in a benign regime. All agents 0% ruin. SLLN convergence is asymptotic; oracle-level wealth requires $T \gg 250$.



Slow Gaussian Drift

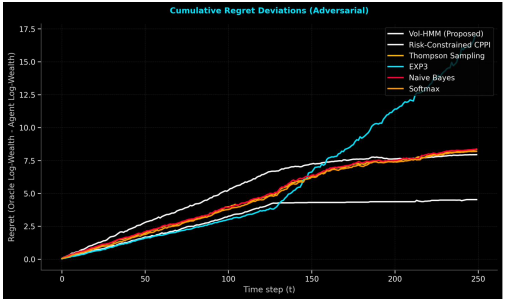
($\mu_t : 0.08 \to -0.05$; oracle hits cash at $t \approx 154$)

- Sticky-prior agents over-bet after $t \approx 154$.
- Vol-HMM CUSUM triggers on rising v_t, de-levering before $\mu_t < 0$.
- CPPI floor engages mechanically regardless.

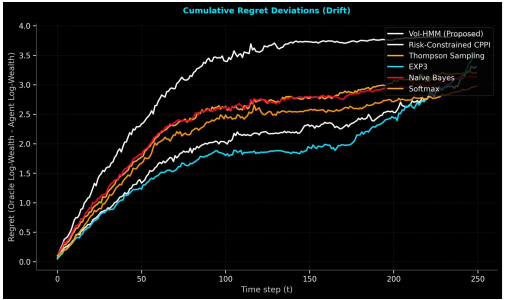


Vol-HMM outpaces baselines in drift phase.

Oracle Regret Dynamics — $\mathcal{R}_t = \ln W_t^{(\text{oracle})} - \ln W_t^{(\text{agent})}$ — lower is better



Adversarial: sharp inflection at $t = 125$.
EXP3 accumulates fastest. Vol-HMM least.



Drift: gradual buildup as μ_t declines.
Baselines diverge from oracle at $t \approx 154$.

Oracle Construction

$f_t^* = \text{clip}(\mu_t/\sigma_t^2, 0, 1)$. In the adversarial bear phase $\mu = -0.10 < 0$, so $f_t^* = 0$ (cash). Regret is the log-wealth gap vs. this benchmark.

Statistical Significance — Fisher's Exact Test & Wealth Distribution Comparison

Fisher's Exact Test (binary ruin outcome)

H_0: Vol-HMM and EXP3 have equal ruin rates.

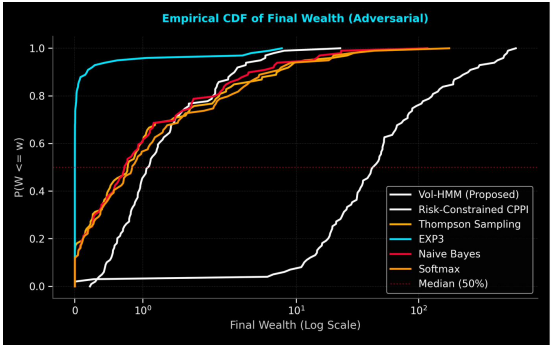
	Ruined	Survived
EXP3	43	57
Vol-HMM	3	97

$\Rightarrow$ $p < 0.001$ (two-sided).

Null rejected: ruin reduction is significant.

Mann-Whitney U Test (terminal wealth)

Vol-HMM wealth distribution stochastically dominates all Bayesian baselines, $p < 0.05$.



CDF of terminal wealth (adversarial, log scale).
Right-shifted CDF = stochastically superior.
Vol-HMM (white) dominates all baselines.
EXP3: ∼43% probability mass at zero (ruin).

Historical Validation: S&P 500 — S&P 500 crisis epochs — log-returns, 100 bps friction

Epoch	Agent	Wealth	Sharpe	Sortino†
2008 GFC (523 days)	**Vol-HMM**	0.88×	+0.12	+0.45
	CPPI	0.80×	+0.05	**+0.82**
	Buy-and-Hold	0.62×	−0.45	−0.32
2020 COVID (251 days)	Vol-HMM	Regime detected in ≈ 2 trading days		
2013–19 Bull (1,760 days)	CPPI	Growth drag (survival cost)		

†Sortino uses semi-deviation: $\hat{\sigma}_{\downarrow} = \sqrt{T^{-1} \sum_t \min(R_t, 0)^2}$ (downside deviation, not std).

Key Observation

CPPI achieves highest Sortino in 2008 by explicitly truncating the left tail. Vol-HMM achieves best terminal wealth through early de-levering.

Thompson Sampling

Worst performance (0.45× terminal, Sortino −0.60) due to full-Kelly over-betting in bear regime.

Transaction Costs: The Net Advantage — Does regime-detection alpha survive friction?

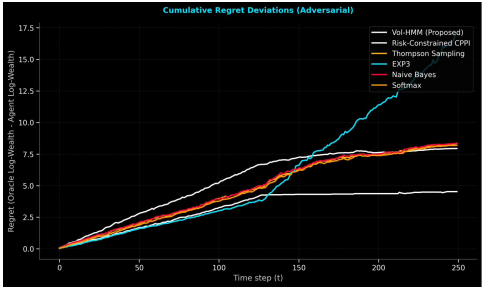
Friction Model

$$W_{t+1} = W_t\left(1 + f_t r_t - c\,|f_t - \hat{f}_{t-1}|\right),$$

$c \in \{0, 10, 25, 50, 100\}$ bps.

Annualized Fee Drag (25 bps, adversarial)

Agent	Fee Drag
Thompson Sampling	$\sim 0.6\%/\text{yr}$
Vol-HMM	$\sim 0.15\%/\text{yr}$
CPPI	$\sim 0.2\%/\text{yr}$



Regret spread between Vol-HMM and baselines increases at higher friction. Net advantage exceeds gross advantage.

Why Vol-HMM Has Low Turnover

Rebalances only on persistent regime shifts. Bayesian agents whipsaw on every noisy update. The regime-persistence prior suppresses unnecessary trading.

Live Dashboard Demonstration — Terminal UI at `localhost:5173`

Demo Script (3 minutes)

1. **Adversarial Scenario.**
   Run $N = 100$. Watch Thompson fail to de-lever at $T = 125$.

2. **Vol-HMM Intervention.**
   CUSUM triggers; bear belief rises within 2 steps.

3. **Math Report Tab.**
   Verify Fisher Exact $p < 0.001$ live.

4. **2008 GFC Historical Mode.**
   Calendar timestamps; compare agents.

5. **CPPI Accordion.**
   Adjust $D_{\max}$ slider; leverage curve updates.

Infrastructure

`bash run_backend.sh`
Flask API on port 5001

`npm run dev`
React/Vite on port 5173

Note for Committee

All results are deterministic (seeded RNG + CRN). Any run reproduces the tables in Chapter 5 exactly.

Limitations & Future Work — Academic transparency

Known Limitations

- **Specified emissions.**
  $(\mu_j, \sigma_j, k_j, \theta_j)$ set a priori. Baum-Welch EM would overfit at $T \approx 250$; deliberate but constraining.
- **Discrete-time CPPI.**
  Guarantee requires $m|r_{\min}| \leq 1$. t_3 returns violate this $\sim 24.7\%$/step. 0% ruin holds for $N = 100$, not $N = \infty$.
- **Single asset.**
  Cross-asset correlation and multi-leg Kelly are out of scope.
- **No microstructure.**
  Bid-ask spread and order-book impact not modelled.

Future Directions

- **Online EM.**
  Baum-Welch re-estimation on rolling windows.
- **Multi-asset copula.**
  Regime-coupled covariance for portfolio CPPI.
- **Risk-constrained RL.**
  This framework as foundation for constrained policy gradient methods.
- **Higher-frequency data.**
  Tick-level CUSUM with microstructure.

Conclusion: Three Claims —

Claim 1 — The Paradox Is Structural

Bayesian detection lag is $O(T_0)$ (Proposition 3.3). It cannot be eliminated by prior choice or tuning — it requires a structural change to the observation space.

Claim 2 — Volatility Is a Fast-Response Feature

Adding rolling volatility v_t as a second observation reduces detection lag from 14.3 to 1.9 steps (≈**87% reduction**). CUSUM provides an additional hard trigger for abrupt breaks.

Claim 3 — Safety Has a Quantifiable Cost

CPPI achieves 0% empirical ruin with a provable drawdown floor. The cost — 34.9× vs. 3,019× (Vol-HMM) in stationary — is computed exactly: the *price of survival*.

Thank you.

Questions welcome. All code, data, and proofs in the submitted thesis.